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January 2027

Ira A. Fulton Schools of Engineering

School of Computing and Augmented Intelligence

Arizona State University

Letter of Intent - Doctor of Engineering in Engineering Management, Spring 2027

My older brother EJ and I grew up in an abusive home. From the time I was a child, I needed to understand something that took me decades to be able to name clearly: why would a parent intentionally harm their own children? That question was not academic. It was survival. It drove me into psychology the way hunger drives you toward food - with urgency, with desperation, and with the understanding that the answer mattered more than anything else. I studied human behavior, trauma, and the architecture of the mind informally for decades before I ever pursued a formal credential. I was trying to understand what had happened to us. And I was trying to figure out how to heal it.

EJ left home before me. The day I escaped, he was not there to protect me. I was 15, in 10th grade, and I had no one to call - no family to go to, no friends, because isolation had been part of how we were controlled. I left with nowhere to go and no one expecting me. I did not know from night to night where I would sleep or whether I would be safe. What fear did to me was push me into education. It became my refuge, my structure, the one thing the chaos could not take. I never dropped out. Not once.

I also walked the path that so many of my future clients would walk. I had been on and off psychiatric medication for years - the conventional route, the one the system offers when you show up asking for help. My last appointment with a psychiatrist came after an extended period where I had fallen off - stopped coming, stopped taking the medication, which the office knew because it was distributed there. When I returned, the doctor noted she had not seen me in a while. I told her I was surprised her office had not reached out. She plainly said follow-up was not her responsibility - despite knowing I had been absent, despite my documented history of suicidal ideation, despite the fact that I was a new mother navigating postpartum depression. She had noticed I was gone. She had decided that noticing was enough.

I never went back. Not to her, and not to that model. What I understood in that moment - and what I have understood more clearly every year since - is that the system is not designed around the patient. It is designed around the appointment. Around the calendar. Around the administrative structure that exists between a person in crisis and the care they need. The gap is not accidental. It is architectural. And I had been living inside it my entire life.

What I understand now is that my mother was also failed by the same system. Her untreated trauma became the source of ours. The system that was supposed to heal her instead left her to damage the people she was supposed to protect. This wasn't personal pathology - this was systemic failure cascading through generations. The hunger I felt to understand 'why' wasn't just about survival - it was about breaking a cycle that had already claimed too many.

I went a different direction. At the Energetic Health Institute I learned about the connection between food and mood - and for the first time encountered the neurotransmitter systems that would later become central to FSL's clinical design. That led me to the Academy for Addiction and Mental Health Nutrition, where I studied amino acid therapies with a specific clinical focus on anxiety, depression, bipolar disorder, and schizophrenia - studying the foundational work of Linus Pauling, Abram Hoffer, William Walsh, and Albert Mensah. I became Board Certified in Holistic Nutrition (BCHN) and kept going. I enrolled at Kingdom College of Natural Health and obtained my Doctor of Naturopathic Psychology (D.N.Psy.) with an orthomolecular focus - applying everything I had learned about biochemistry, amino acid intervention, and nutritional psychiatry in a formal doctoral context. I then completed Medical Neuroscience at Duke University School of Medicine. The more I learned, the more I needed to know. One thing led to the next - driven by genuine curiosity that never stopped.

EJ and I carried what happened to us in that house long after we left. We understood each other in a way only people who survived the same thing can. He made it to 50. He got a promotion. He felt overwhelmed and did what I had done - he asked for help. He went to his doctor. He was put on mental health medication. He noticed it did not feel right and said so. He called the office. He made an appointment. The office cancelled it. And cancelled it again. And cancelled it a third time. While he was on medication that was altering his neurochemistry, in a window of acute vulnerability that required urgency, the system sent him a cancellation notice. Three times.

On October 29, 2023 - six days after his 50th birthday - EJ shot himself.

I had sat in that psychiatrist's chair and been told my absence was not her responsibility. EJ had made three appointments and been turned away every time. We came from the same house. We fought the same battles. We both reached for the same system. The difference in our outcomes was not effort. It was not willingness. It was not strength. It was a cancelled appointment. It was a doctor who decided follow-up was not her job.

EJ did not fail the system. The system failed EJ. Specifically, precisely, and preventably. But EJ's death made it clear that individual healing wasn't enough. The gaps that failed my mother, that failed me, that killed my brother - these aren't edge cases. They're architectural features of a system designed around institutional convenience rather than human need. I built Future Systems Lab not just for people like me and EJ, but for every family caught in cycles the current system perpetuates rather than breaks. Clinical work can only ever reach one person at a time. A system is unlimited. A system can outlive me. People like my family deserve a response at that scale. I believe the universe placed me in charge of building it because I am relentless in purpose - and I have the lived experience, the clinical foundation, and now the technical infrastructure to see it through.

The Practitioner: Building the Clinical Foundation

For thirty years I worked in clinical care, including as a licensed dental provider. What I learned across those decades is that the dental chair is also a counseling chair. More people than not - even before I had my psychology degree - would open up to me about heavy psychological and emotional disruptions. Grief, anxiety, trauma, the weight of lives people were barely holding together. It is more common than the healthcare system acknowledges, and the connection is not incidental: psychological distress is one of the most consistent drivers of oral health neglect. The mouth tells you what the mind has not yet been given space to say. I listened. And what I kept seeing was the same pattern: people reaching for help and finding a system that was not designed to meet them.

I opened a naturopathic psychology center. I redesigned the interior of the space myself - because environment matters to healing, and I knew from my own life what it felt like to be in spaces that did not hold me safely. I developed a CBD wellness line from seed to shelf. I was building an ecosystem of care, not just a practice. Then the pandemic hit. I transitioned online and implemented crypto payments - my first step toward understanding that the financial infrastructure of healthcare was part of the access problem. And then EJ died.

After losing him I enrolled in Johns Hopkins University's Health Informatics program and was awarded a Specialization Certificate. For the first time I had language for what I had been observing across thirty years of clinical work. Healthcare was not just a human problem. It was an information systems problem. I went further - completing Health Informatics on FHIR at Georgia Institute of Technology, and an INSEAD specialization in Web3 and Blockchain in Global Commerce. I became a Certified Blockchain and Healthcare Professional (CBHP), a Certified Smart Contract Auditor, and earned the SSI Master designation in Self-Sovereign Identity. I completed certifications in IPFS, Filecoin, Merkle DAGs, decentralized content addressing, and libp2p. I added IBM Business Analyst and Google Data Analytics credentials. Within six months of EJ's death I was teaching myself Web3, decentralized systems, and systems engineering from scratch. No prior computer experience. None. What I had was the same relentlessness that had kept me in school when I had nowhere to sleep. I watched the sun go down and come back up - regularly, for months - determined to figure it out. And I did.

Basic Qualifications

Academic credentials include: D.N.Psy. with an orthomolecular focus (Kingdom College of Natural Health); BS in Management Information Systems (Colorado State University Global, HLC accredited); Health Informatics Specialization Certificate (Johns Hopkins University); Web3 and Blockchain in Global Commerce (INSEAD); Medical Neuroscience (Duke University School of Medicine); Health Informatics on FHIR (Georgia Institute of Technology).

Provider credentials include National Provider Identifier (NPI) 1497696264, issued by NPPES, Taxonomy 175F00000X (Naturopath, Primary), reflecting active licensure as a credentialed healthcare provider.

My career spans nearly three decades of clinical and entrepreneurial leadership. I began in 1996 as a dental assistant in California, where I was trained in expanded functions including IV sedation placement and monitoring, local anesthesia administration, and radiology. From 1998 to 2002, I worked as a contract dental assistant across NC practices addressing shortages. In 2002, I graduated with an AAS in Dental Hygiene from Coastal Carolina Community College and have since maintained ongoing multi-practice contract work across North Carolina.

The defining phase of my clinical career began in 2017 when I contracted with Riccobene Associates, initially as one of my multiple NC practice engagements. In 2019, when pandemic-driven hygienist shortages threatened throughput across Riccobene's 70+ office network, CEO Michael Riccobene personally recruited me for a company-wide engineering intervention. Working directly with Riccobene's C-suite, I designed and implemented a two-tier production-share compensation model: a 27% standard tier that was subsequently adopted across all company locations, and a 33% architect tier for myself as the system designer. This required restructuring workflow processes across multiple office locations while implementing the model network-wide.

The results were measurable and sustained. Documented per-provider production across the network grew from $380,710 in 2021 to $624,640 annualized through October 2024 — a 64% increase across four documented years of continuous year-over-year growth. (Documented figures begin 2021 following a practice management software migration; pre-migration data from the model’s 2019 launch is not retrievable in standardized format.). Concurrently, the hygienist share of total office production grew from 30.4% to 38.1%, demonstrating that the system redesign enhanced overall office economics rather than simply redistributing existing revenue. These metrics positioned my compensation at approximately 1.5-2× the North Carolina state benchmark for practitioners in the role. Throughout this 6-year engagement with Riccobene (2019-2025), I maintained ownership and operations of my wellness businesses, demonstrating the intersection of clinical excellence and entrepreneurial leadership that would later inform FSL's design.

Since June 2025, I have returned to multi-practice contract work, now providing both active clinical hygiene services and compensation model consulting to NC practices implementing structurally similar frameworks during the ongoing statewide shortage.

I founded SunCodes Holistic Health (2018-2023), and launched both Future Systems Lab and the Naturopathic Psychology and Hypnosis Center in 2023. The attached CV documents the full record.

I would welcome the opportunity to discuss my qualification pathway directly with the admissions committee if any clarification would be helpful.

Why I Built Future Systems Lab: Five Problems, One Ecosystem

I built Future Systems Lab to answer one engineering question: what does a data sovereignty system look like when it is designed around the person — their access, their autonomy, their anonymity — instead of around institutional convenience? I chose mental health as the proving ground because it is the hardest possible test. If the architecture holds for behavioral health — where stigma is highest, consent is most complex, and the human cost of failure is measured in lives — it holds for every other domain where trust, access, and sovereignty are non-negotiable design requirements.

The five problems I set out to solve are the same five that made EJ's path so hard:

Stigma - FSL does not use diagnostic language. HypnoNeuro's protocol is organized around neurotransmitter systems (serotonin, dopamine, GABA, endorphins, endocannabinoid) rather than diagnostic categories. A person selects what they want to work on, not what label applies to them.

Shame - The platform is wallet-gated. There is no username tied to an email address, no insurance card, no referral form. You connect with your wallet and you are sovereign from the moment you arrive.

Access - Decentralized architecture removes geographic and insurance gatekeeping. A person in rural North Carolina and a person in São Paulo have the same front door.

Privacy - A consent-first architecture means your health data does not move without your cryptographic permission. SovereignLedger is the governance layer that ensures this is not a policy promise - it is an on-chain guarantee.

Autonomy - Participants own their session history, their wellness journey, and their earned tokens. The AlchemistForge module allows users to record personal transformation on-chain. This is not a metaphor. It is a technical commitment.

FSL is live. It is not a prototype, a pitch deck, or a grant proposal. It is a deployed, multi-platform Web3 decentralized infrastructure for sovereign data governance encompassing five infrastructure components: EncryptHealth (unified entry point with wallet-based authentication), HypnoNeuro (hypnotherapy and neurotransmitter-based wellness sessions with 45 browser-based therapeutic games (15 per level across L1 Hypnosis, L2 Orthomolecular, and L3 Inner Child)), SovereignLedger (health data governance infrastructure), AlchemistForge (shadow work and on-chain verified personal transformation), and NeuroBalance (backend wellness intelligence layer). Eight smart contracts are live on Ethereum Sepolia testnet. A 17-agent AI council runs 24/7 on self-hosted infrastructure. A USPTO trademark application is on file (Serial No. 99533250). The platform has been built, governed, and managed entirely under my technical and strategic leadership.

Behavioral health was chosen as the proving ground because it imposes the strictest regulatory load and the highest stigma cost of any deployable domain. More personally: naturopathic psychology — my clinical practice — has never had infrastructure that scales. FSL is the first time someone can experience that model without being in a clinical setting. Building the architecture that makes that possible, at scale, with sovereignty as the design constraint, is the applied research contribution.

Why Arizona State University - and Who I Want to Work With

I did not arrive at ASU by default. I came looking for three specific people whose work maps directly onto the three pillars of FSL's architecture.

Dr. Dragan Boscovic (ASU Blockchain Research Lab - School of Computing and Augmented Intelligence) is the primary advisor I would want. His lab's research direction is the most direct intellectual ancestor of what FSL has built. Since 2019 his Blockchain Research Lab has been advancing exactly what FSL implements: patient-controlled medical data, consent-first sharing, blockchain healthcare infrastructure. His former graduate student Manish Vishnoi's master's thesis - "How people can maintain ownership of their medical data and only share it on a need-to-know basis" - is FSL's literature foundation. His three-component blockchain framework (scalability, security, decentralization) captures the design tradeoffs FSL navigated. His expertise in tokenomics, smart contract verification, and trustless information systems for health data would bring exactly the formal rigor this applied project needs to become publishable, fundable research.

Dr. Gail-Joon Ahn (School of Computing and Augmented Intelligence - SEFCOM Laboratory) is the security and privacy co-advisor this project demands. IEEE Fellow. Director of ASU's Center for Cybersecurity and Trusted Foundations. His research foci - identity management, access control architecture for distributed systems, privacy-aware collaborative data sharing, and formal models for computer security - are the exact theoretical foundation for FSL's consent-gate protocol, wallet-gated authentication architecture, and SovereignLedger governance model. Every security decision I have made in the FSL stack was made against the backdrop of exactly the problems his lab has spent decades formalizing. I want that formalization applied to what I have built.

Dr. Hassan Ghasemzadeh (College of Health Solutions - Biomedical Informatics & Data Science) is the clinical informatics co-advisor who validates the proving ground. His lab's work on digital health systems, health data analytics, clinical informatics, and AI-driven behavioral health intervention sits precisely at the intersection of what FSL is doing clinically and what the DEng applied project needs to formalize. His research on counterfactual modeling for health intervention design, wearable biosensors for real-world behavioral monitoring, and personalized treatment adherence validates that FSL's behavioral health framing is clinically rigorous. I want to bring a live system into his research environment and do the work of formalizing it.

Together, these three represent the complete advisory architecture - with Dr. Boscovic anchoring the blockchain engineering pillar, Dr. Ahn formalizing the security and identity foundations, and Dr. Ghasemzadeh validating the clinical depth of the behavioral health proving ground.

The Applied Project: On-Chain Claims Lifecycle Ecosystem for Sovereign Behavioral Health

My proposed applied project is the formal engineering management documentation, validation, and scalability framework for what Future Systems Lab has built: a decentralized, consent-first, blockchain-gated behavioral health delivery ecosystem designed to serve populations for whom the current system has failed.

The project will address four engineering research questions. First: what is the optimal architecture for a consent-first data governance system that achieves regulatory-grade privacy standards without centralized data storage — using HIPAA-regulated behavioral health as the highest-bar test case — though FSL itself holds no Protected Health Information by architectural design for a generalizable design pattern? Second: how do tokenized incentive structures - specifically earned-only token models, NFT-based access tiers, and community contribution mechanisms - affect Sovereign Guide participation and sustained participant engagement in decentralized infrastructure platforms? Third: what are the engineering management trade-offs between decentralized and centralized governance models for health data platforms serving vulnerable populations? Fourth: what does a replicable deployment framework look like for sovereign health infrastructure in medically underserved communities?

The platform to be studied already exists and is operational. The DEng applied project produces what comes next: the peer-reviewed documentation, the formal engineering analysis, the scalability model, and the institutional-grade research output that positions this infrastructure for NIH grant applications, academic partnerships, and community deployment beyond the current platform footprint.

This is not a theoretical exercise. It is applied engineering research in the most literal sense - research that begins with a system already in production and asks: how do we make it rigorous enough, documented enough, and replicable enough to reach everyone who needs it?

The 45 therapeutic engagement sessions currently deployed across HypnoNeuro's three protocol layers (L1 GABA/endocannabinoid, L2 orthomolecular, L3 inner child healing) are functional infrastructure scaffolds — the wallet gating, HNT minting on completion, and on-chain achievement system are fully operational. The DEng applied research contribution is the protocol depth layer: EMDR, EFT, somatic visualization, and AI-adaptive wearable-integrated protocols designed in collaboration with faculty whose research domains directly map to FSL's therapeutic architecture. Dr. Hassan Ghasemzadeh's work in AI-powered wearable sensors and orthomolecular AI refinement represents the precise intersection where FSL's infrastructure meets the clinical depth it was designed to carry. The current sessions are not the research — they are the proving ground the research will run on.

SovereignSession Phases 1–4 establish the technical foundation for a sovereign wellness metaverse: individual therapeutic sessions converging with group support spaces, all avatar-mediated, all wallet-native, all pseudonymous. Phase 5 (encrypted recording with participant-controlled keys) and the subsequent avatar layer represent the frontier engineering problems the DEng research program is designed to address — problems that require both doctoral-level rigor and the applied infrastructure already deployed to solve them.

FSL is, at its core, an applied engineering thesis: that the centralized data architecture underlying nearly every modern institution — healthcare, finance, government, education, identity — is structurally broken in the same way, and that a consent-first, wallet-gated, decentralized governance pattern resolves all of it. Mental health is where I have stress-tested this thesis because it is the domain where centralized failure costs lives most directly. The documentation, scalability framework, and validated reference architecture this applied project produces will be applicable to any system where access, autonomy, anonymity, and trustless governance are design requirements — and the engineering management discipline of formalizing that pattern is precisely what a Doctor of Engineering is for.

I welcome the opportunity to work with Arizona State University and am deeply grateful for the consideration. I have been invited on several occasions to speak at TEDx and have declined - because the timing was not right and the platform needed to be earned, not just offered. It is my intention to accept that invitation as an ASU graduate student, with the research and the depth of understanding to stand behind every word. I am looking to align with people who share this vision - to help scale and bring this ecosystem to those who need it most: people like me and EJ.

Warm regards,

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AI Use Disclosure

```
// SPDX-License-Identifier: Sovereign
pragma sovereignty ^1.0.0;

contract AIDisclosure {
    address public author = MegMontanezDavenport;
    string[] public tools = ["Claude (Anthropic)"];
    string public intent = "Drafting assistance, structure refinement";

    function ownership() public pure returns (string memory) {
        return "Ideas, lived experience, and framing: 100% human.";
    }
}
```

Listing 1: AI Use Disclosure